Chapter 16, Problem 10P

Problem

The step responses of a series *RLC* circuit are

$V_c = 40 - 10e^{-2000t} - 10e^{-4000t}$ V, $t > 0$

$i_L(t) = 3e^{-2000t} + 6e^{-4000t}$ mA, $t > 0$

(a) Find *C*.

(b) Determine what type of damping is exhibited by the circuit.

Step-by-step solution

Step 1 of 4

Consider the responses of series *RLC* circuit is

$$i_L(t) = \left(3e^{-2000t} + 6e^{-4000t}\right) \text{ mA}$$

$$v_C(t) = \left(40 - 10e^{-2000t} - 10e^{-4000t}\right) \text{ V}$$

Draw the series *RLC* circuit.

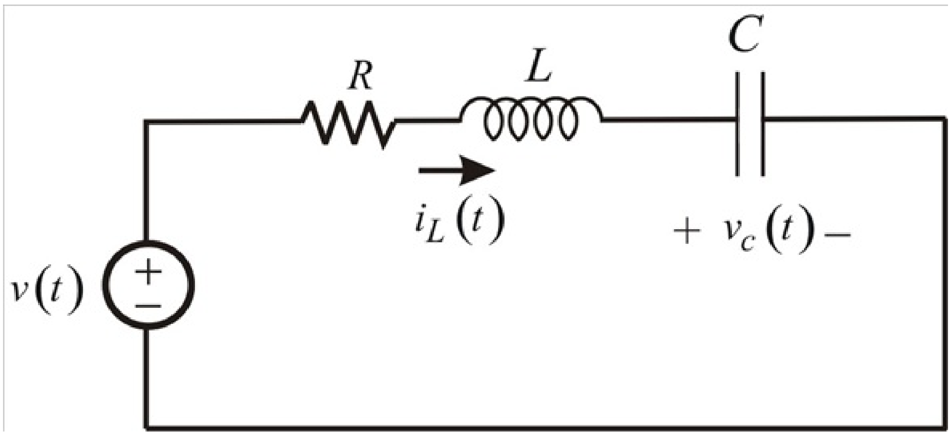

Figure 1: Series *RLC* circuit

Step 2 of 4

(a)

In series *RLC* circuit same current flows through the each element. So,

$$i_L(t) = i_c(t)$$

$$i_L(t) = C\frac{dv_C(t)}{dt}$$

$$\left(3e^{-2000t} + 6e^{-4000t}\right) \times 10^{-3} = C\frac{d}{dt}\left(40 - 10e^{-2000t} - 10e^{-4000t}\right)$$

$$\left(3e^{-2000t} + 6e^{-4000t}\right) \times 10^{-3} = C\left(2\times10^4 e^{-2000t} + 4\times10^4 e^{-4000t}\right)$$

Compare the exponential terms.

$$3\times10^{-3} = 2\times10^4 C$$

$$C = 0.15\times10^{-6}$$

$$C = 0.15\,\mu\text{F}$$

Thus, the capacitor value is $\boxed{0.15\,\mu\text{F}}$.

(b)

Write the Laplace transform pairs those are used to find $I_L(s)$.

$$u(t) \xleftrightarrow{L} \frac{1}{s}$$

$$e^{-at}u(t) \xleftrightarrow{L} \frac{1}{s+a}$$

Calculate the Laplace transform of $i_L(t)$.

$$\begin{aligned} I_L(s) &= \left(\frac{3}{s+2000} + \frac{6}{s+4000}\right)10^{-3} \\ &= \left[\frac{3(s+4000)+6(s+2000)}{(s+2000)(s+4000)}\right]10^{-3} \\ &= \left[\frac{9s+24000}{(s+2000)(s+4000)}\right]10^{-3} \\ &= \left[\frac{9s+24000}{s^2+6000s+8\times10^6}\right]10^{-3} \end{aligned}$$

Characteristic equation of $I_L(s)$.

$$s^2+6000s+8\times10^6=0$$

Compare the characteristic equation $s^2+6000s+8\times10^6=0$ with the standard equation $s^2+2\zeta\omega_n+\omega_n^2=0$.

$$\omega_n^2 = 8\times10^6 \qquad\qquad 2\zeta\omega_n = 6000$$

$$\omega_n = 2.828\times10^3 \qquad\qquad 2\zeta\left((2.828)(10^3)\right) = 6000$$

$$\zeta = \frac{6000}{(2)(2.828)(10^3)}$$

$$\zeta = 1.06$$

The damping value is greater than 1 $(\zeta > 1)$. So, the system is over damped system.